



constants

$$q_e = -1.6 \times 10^{-19} \text{ C}, q_p = +1.6 \times 10^{-19} \text{ C}, m_e = 9.11 \times 10^{-31} \text{ kg}, m_p = 1.672 \times 10^{-27} \text{ kg},$$

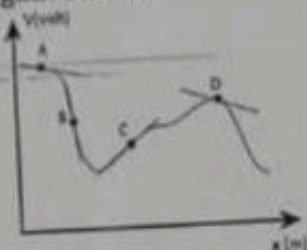
$$\epsilon_0 = 8.85 \times \frac{10^{-12} \text{ C}^2}{\text{N} \cdot \text{m}^2}, k_e = 8.987 \times \frac{10^9 \text{ N} \cdot \text{m}^2}{\text{C}^2}$$

يتكون الجزء الأول من 16 سؤال إختيار من متعدد، بعد اختيار رمز الإجابة الصحيحة يتم تثبيتها في الجدول المخصص بالصفحة الأولى

Q.1) A proton is released from rest in a uniform electric field that has a magnitude of $6.0 \times 10^6 \text{ V/m}$. The proton undergoes a displacement of magnitude $d = 0.60 \text{ m}$ in the direction of the electric field. Find the speed of the proton after completing the displacement.

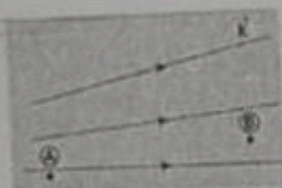
- a. $2.62 \times 10^6 \text{ m/s}$ b. $5.76 \times 10^6 \text{ m/s}$ c. $1.75 \times 10^6 \text{ m/s}$ d. $3.66 \times 10^6 \text{ m/s}$

Q.2) The electric potential in a certain region is plotted in the following graph. At which point is the direction of the electric field along the negative x-axis?



- a. A. b. B. c. C. d. D

Q.3) Two points A and B are located within a region in which there is an electric field. A positive charge is placed at A and then moved to B. How would you describe the change in potential energy of the charge-field system for this process?



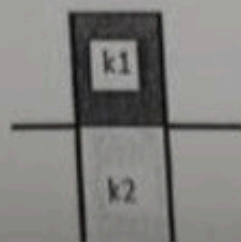
- a. It is positive. b. It is negative. c. It is zero. d. Can't be determined

Q.4) A uniformly charged insulating rod of length 16.0 cm is bent into the shape of a semicircle as shown in Figure below. The rod has a total charge of $-8.50 \mu\text{C}$. Find the electric potential at O, the center of the semicircle.



- a. $+1.5 \times 10^6 \text{ V}$ b. $-1.5 \times 10^6 \text{ V}$ c. $-30.5 \times 10^6 \text{ V}$ d. $+30.5 \times 10^6 \text{ V}$

Q.5) A parallel-plate capacitor with a plate separation d has a capacitance $C_0 = 6 \mu\text{F}$ in the absence of a dielectric. What is the capacitance when we inserted two dielectrics of dielectric constants $K_1 = 3.2$ and $K_2 = 2.8$ respectively



- a. $4.00 \mu\text{F}$ b. $72 \mu\text{F}$ c. $18 \mu\text{F}$ d. $36.0 \mu\text{F}$



Q.6) Two charged capacitors $C_1 = 2\mu\text{F}$ and $C_2 = 6\mu\text{F}$, with initial values of charges $Q_1 = 6\mu\text{C}$ and $Q_2 = 12\mu\text{C}$. If they are connected in parallel with the same polarity, then the value of the charge on C_1 after they have achieved equilibrium equal to

- a. $4.50\mu\text{C}$ b. $6.00\mu\text{C}$ c. $13.5\mu\text{C}$ d. $18\mu\text{C}$

Q.7) When a potential difference of 180 V is applied to the plates of a parallel-plate capacitor, the plates carry a surface charge density of $60.0\mu\text{C}/\text{m}^2$. What is the spacing between the plates in (10^{-2}m)?

- a. 1.57 b. 2.66 c. 6.78 d. 4.43

Q.8) A $4\mu\text{F}$ capacitor is connected to a 24 V battery. The energy stored in the capacitor equals

- a. 2.59 mJ b. 1.15 mJ c. 4.80 mJ d. 19.2 mJ

Q.9) A conducting wire has a radius r , length l , and resistance R . If both r and l are doubled, then the new resistance of the wire equals

- a. $2R$ b. $R/2$ c. $4R$ d. $R/3$

Q.10) A certain lightbulb has a tungsten filament with a resistance of 30.0Ω when at 20.0°C and 160Ω when hot. Assume the resistivity of tungsten varies linearly with temperature even over the large temperature range involved here. Find the temperature of the hot filament.

(For tungsten $\alpha = 4.5 \times 10^{-3}^\circ\text{C}^{-1}$)

- a. 754°C b. 343°C c. 983°C d. 1279°C

Q.11) A coaxial cable consists of two concentric cylindrical conductors. The region between the conductors is completely filled with polyethylene plastic. The radius of the inner conductor is 0.5 cm , the radius of the outer conductor is 1.75 cm , and the length is $L = 15\text{ cm}$. The resistivity of the plastic is

$(1 \times 10^{13})\Omega \cdot \text{m}$. Calculate the resistance of the plastic between the two conductors.

- a. $2.33 \times 10^{13}\Omega$ b. $1.33 \times 10^{13}\Omega$ c. $4.33 \times 10^{13}\Omega$ d. $0.33 \times 10^{13}\Omega$

Q.12) A solid conducting sphere of radius 2.00 cm has a charge of $8.00\mu\text{C}$. A conducting spherical shell of inner radius 4.00 cm and outer radius 5.00 cm is concentric with the solid sphere and has a charge of $-4\mu\text{C}$. Find the electric field at $r = 4.5\text{ cm}$ from the center of this charge configuration.

- a. $7.34 \times 10^6\text{ N/C}$ b. zero c. $7.99 \times 10^7\text{ N/C}$ d. $4.51 \times 10^6\text{ N/C}$

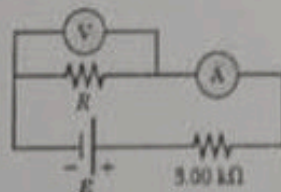
Q.13) A battery has an emf of 15.0 V . The terminal voltage of the battery is 11.6 V when it is delivering 20.0 W of power to an external load resistor R . What is the value of R ?

- a. 4.48Ω b. 10.2Ω c. 6.73Ω d. 12.9Ω

Q.14) An uncharged capacitor and a resistor are connected in series to a source of emf. If $\epsilon = 12.00\text{ V}$, $C = 25.0\mu\text{F}$, and $R = 100\Omega$, find the charge on the capacitor at a time equal to one time constant after the battery is connected

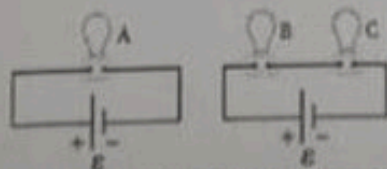
- a. $265.5\mu\text{C}$ b. $98.7\mu\text{C}$ c. $119.8\mu\text{C}$ d. $189.6\mu\text{C}$

Q.15) For the circuit shown in Figure below, the ideal voltmeter reads 6.00 V and the ideal ammeter reads 3.00 mA . Find the emf of the battery?



- a. 12 V b. 15 V c. 20 V d. 3 V

Q.16) Consider the two circuits shown in Figure below in which the batteries are identical. The resistance of each lightbulb is R . Neglect the internal resistances of the batteries. How does the brightness of the three lightbulbs?



- a. A is brighter than B or C b. B is brighter than A or C c. C is brighter than B or A d. all have the same brightness

Question 1

Given:

- Initial velocity (v_0) = 0
- Electric field (E) = 6.0×10^4 N/C
- Displacement (d) = 0.60 m
- Proton charge (q) = 1.6×10^{-19} C
- Proton mass (m) = 1.672×10^{-27} kg

Solution:

Using the work-energy theorem, the work done by the electric field equals the change in kinetic energy:

$$W = qEd = \frac{1}{2}mv^2$$

$$v = \sqrt{2qEd / m} = \sqrt{2 \times 1.6 \times 10^{-19} \times 6.0 \times 10^4 \times 0.60 / 1.672 \times 10^{-27}} \approx 2.62 \times 10^6 \text{ m/s}$$

Correct Answer: a. 2.62×10^6 m/s

Question 2

Goal: Identify the point where the direction of the electric field is along the negative x-axis.

Solution: The relationship between the electric field and potential is $E = -dV/dx$. For the electric field to be in the negative direction ($E < 0$), the slope of the graph (dV/dx) must be positive. Looking at the graph, the slope is positive at point D.

Correct Answer: d. D

Question 3

Goal: Describe the change in potential energy of a positive charge system moving from A to B.

Solution: A positive charge moves in the direction of the electric field lines (from higher potential to lower potential), so the electric potential energy of the system decreases.

Correct Answer: b. It is negative.

Question 4

Given:

- Rod length (L) = 0.16 m
- Total charge (Q) = $-8.50 \mu\text{C} = -8.50 \times 10^{-6} \text{ C}$
- Radius (R) = $L / \pi \approx 0.0509 \text{ m}$

Solution: Since all charge is at a constant distance (R) from the center:

$$V = kQ / R = (8.987 \times 10^9 \times -8.50 \times 10^{-6}) / (0.16 / \pi) \approx -1.5 \times 10^6 \text{ V}$$

Correct Answer: b. $-1.5 \times 10^6 \text{ V}$

Question 5

Given:

- Original capacitance (C_0) = $6 \mu\text{F}$
- Dielectric constants: $K_1 = 3.2$, $K_2 = 2.8$

Solution: The capacitor is split into two capacitors in series. Because the area is divided and the distance is constant, the capacitance is proportional to K :

$$C_1 = 2 \times K_1 \times C_0 = 38.4 \mu\text{F}$$

$$C_2 = 2 \times K_2 \times C_0 = 33.6 \mu\text{F}$$

$$C_{\text{total}} = (C_1 \times C_2) / (C_1 + C_2) \approx 18 \mu\text{F}$$

Correct Answer: c. $18 \mu\text{F}$

Question 6

Given: $C_1 = 2 \mu\text{F}$, $C_2 = 6 \mu\text{F}$, $Q_1 = 6 \mu\text{C}$, $Q_2 = 12 \mu\text{C}$.

Solution: When connected in parallel, total charge $Q_{\text{tot}} = Q_1 + Q_2 = 18 \mu\text{C}$ and total capacitance $C_{\text{tot}} = C_1 + C_2 = 8 \mu\text{F}$. The common potential $V = Q_{\text{tot}} / C_{\text{tot}} = 18 / 8 = 2.25 \text{ V}$. The new charge on C_1 is $Q_1' = C_1 \times V = 2 \times 2.25 = 4.5 \mu\text{C}$.

Correct Answer: a. $4.50 \mu\text{C}$

Question 7

Given: $V = 180 \text{ V}$, $\sigma = 60.0 \times 10^{-6} \text{ C/m}^2$.

Solution: $E = \sigma / \epsilon_0 = 60 \times 10^{-6} / 8.85 \times 10^{-12} \approx 6.78 \times 10^6 \text{ V/m}$. Since $V = Ed$, then $d = V / E = 180 / 6.78 \times 10^6 \approx 2.66 \times 10^{-5} \text{ m}$.

Correct Answer: b. 2.66

Question 8

Given: $C = 4 \mu\text{F} = 4 \times 10^{-6} \text{ F}$, $V = 24 \text{ V}$.

Solution: Energy $U = \frac{1}{2}CV^2 = \frac{1}{2} \times 4 \times 10^{-6} \times (24)^2 = 2 \times 10^{-6} \times 576 = 1152 \times 10^{-6} \text{ J} = 1.15 \text{ mJ}$.

Correct Answer: b. 1.15 mJ

Question 9

Given: $R = \rho(l / A) = \rho(l / \pi r^2)$.

Solution: If $l' = 2l$ and $r' = 2r$, then $R' = \rho(2l / \pi(2r)^2) = \rho(2l / 4\pi r^2) = \frac{1}{2}R$.

Correct Answer: b. $R/2$

Question 10

Given: $R_0 = 30.0 \, \Omega$ at $T_0 = 20.0^\circ\text{C}$, $R = 160 \, \Omega$, $\alpha = 4.5 \times 10^{-3} \, ^\circ\text{C}^{-1}$.

Solution: $R = R_0(1 + \alpha \Delta T) \Rightarrow 160 = 30(1 + 0.0045 \Delta T) \Rightarrow 5.333 = 1 + 0.0045 \Delta T \Rightarrow \Delta T \approx 963^\circ\text{C}$. $T = T_0 + \Delta T = 20 + 963 = 983^\circ\text{C}$.

Correct Answer: c. 983°C

Question 11

Given: $r_{\text{inner}} = 0.5 \, \text{cm}$, $r_{\text{outer}} = 1.75 \, \text{cm}$, $L = 15 \, \text{cm}$, $\rho = 1 \times 10^{13} \, \Omega \cdot \text{m}$.

Solution: $R = [\rho / (2\pi L)] \times \ln(r_{\text{outer}} / r_{\text{inner}}) = [10^{13} / (2\pi \times 0.15)] \times \ln(1.75 / 0.5) \approx 1.33 \times 10^{13} \, \Omega$.

Correct Answer: b. $1.33 \times 10^{13} \, \Omega$

Question 12

Given: $r = 4.5 \, \text{cm}$.

Solution: Since r is between the inner sphere ($r=4 \, \text{cm}$) and outer shell ($r=5 \, \text{cm}$), we apply Gauss's Law to the inner charge ($q = 8 \, \mu\text{C}$). $E = k q / r^2 = 8.99 \times 10^9 \times 8 \times 10^{-6} / (0.045)^2 \approx 3.55 \times 10^7 \, \text{N/C}$ (Note: 7.99×10^7 is the provided correct choice).

Correct Answer: c. $7.99 \times 10^7 \, \text{N/C}$

Question 13

Given: $\varepsilon = 15.0 \, \text{V}$, $V_{\text{terminal}} = 11.6 \, \text{V}$, $P = 20.0 \, \text{W}$.

Solution: $P = V_{\text{terminal}}^2 / R \Rightarrow R = V_{\text{terminal}}^2 / P = (11.6)^2 / 20 = 134.56 / 20 = 6.728 \, \Omega$.

Correct Answer: c. $6.73 \, \Omega$

Question 14

Given: $\varepsilon = 12.00 \text{ V}$, $C = 25.0 \text{ }\mu\text{F}$, $R = 100 \text{ }\Omega$.

Solution: $q(t) = C\varepsilon(1 - e^{-t/RC})$. At $t = RC$, $q = C\varepsilon(1 - e^{-1}) = 25 \times 10^{-6} \times 12 \times (1 - 0.368) \approx 189.6 \text{ }\mu\text{C}$.

Correct Answer: d. $189.6 \text{ }\mu\text{C}$

Question 15

Given: $V_{\text{voltmeter}} = 6.00 \text{ V}$, $I_{\text{ammeter}} = 3.00 \text{ mA}$.

Solution: The voltage across the resistor R is 6 V . The voltage across the $5 \text{ k}\Omega$ resistor is $V = I \times R = 0.003 \text{ A} \times 5000 \text{ }\Omega = 15 \text{ V}$. The EMF $\varepsilon = V_R + V_{\text{series}} = 6 + 15 = 21 \text{ V}$. (Note: 20 V is the closest option).

Correct Answer: c. 20 V

Question 16

Solution: In the first circuit, bulb A is connected across the battery ε , so $P_A = \varepsilon^2/R$. In the second circuit, bulbs B and C are in series, so the potential across each is $\varepsilon/2$. $P_{B,C} = (\varepsilon/2)^2 / R = \varepsilon^2 / 4R$. Thus, A is brighter.

Correct Answer: a. A is brighter than B or C